# Assignment 5: Data Visualization

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## OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

## Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

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## Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1 Loading packages and data
library(tidyverse);library(here);library(cowplot)
library(ggplot2);library(ggthemes)
getwd()
```

```
## [1] "/home/guest/872L/EDE_Spring2026"
```

```
key.data <- here("Data/Processed_KEY")
PeterPaul.chemistry <- read.csv(
  here(key.data,"NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE)

Litter <- read.csv(
```

```r
  here(key.data,"NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE)

#2 Converting factors to date
class(PeterPaul.chemistry$sampledate) #factor
```

```
## [1] "factor"
```

```r
class(Litter$collectDate) #factor
```

```
## [1] "factor"
```

```r
PeterPaul.chemistry$sampledate <- ymd(PeterPaul.chemistry$sampledate)

Litter$collectDate <- ymd(Litter$collectDate)
```

## Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
#3 Creating a default theme
default.theme <- theme_light() +
  theme(axis.text = element_text(color = "black", face = "bold"),
        axis.title = element_text(face = "bold"),
        legend.box.background = element_rect(linewidth = 1),
        legend.position = "right")
```

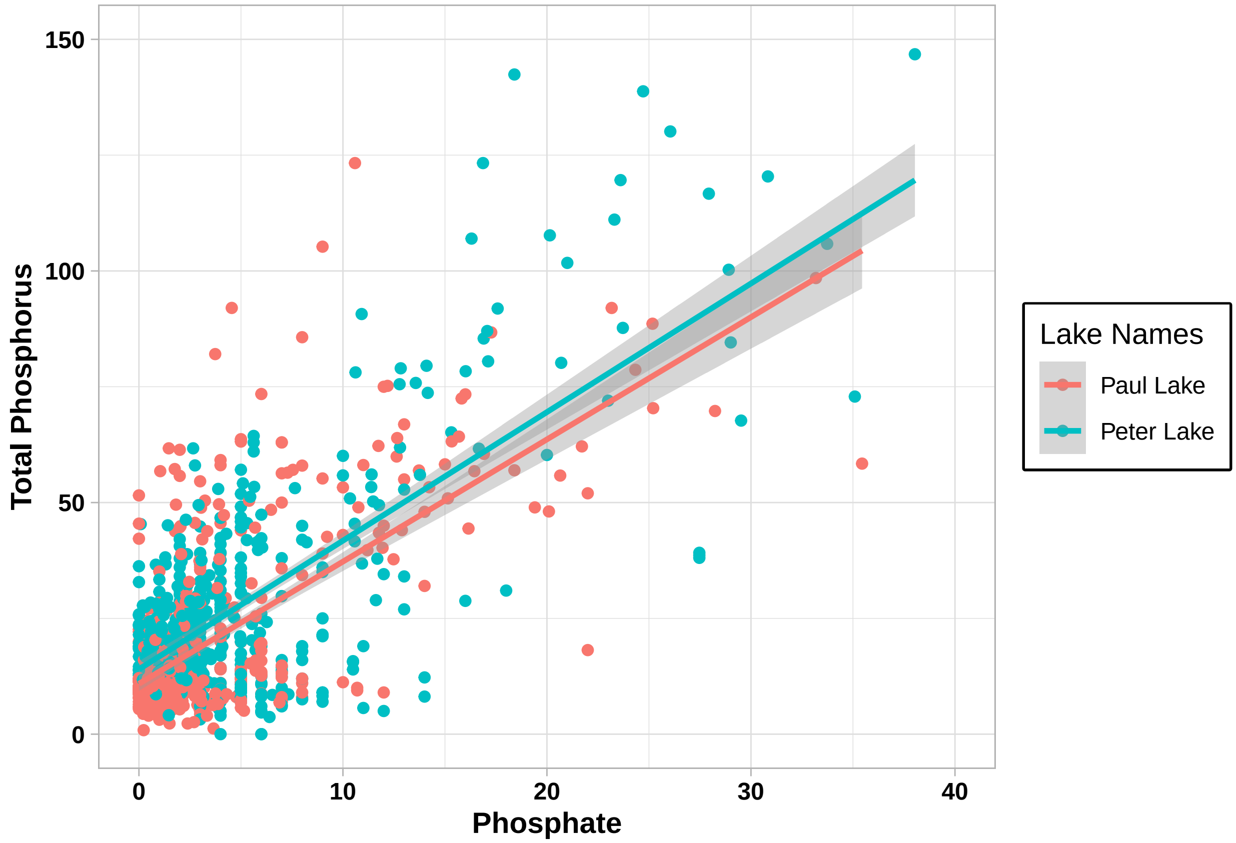
## Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```r
#4 Creating a scatterplot

ggplot(PeterPaul.chemistry) +
  geom_point(aes(y = tp_ug, x = po4, color = lakename)) +
  geom_smooth(method = lm, aes(y = tp_ug, x = po4, color = lakename)) +
  labs(x="Phosphate",y="Total Phosphorus",color = "Lake Names") +
  xlim(0,40) +
  ylim(0,150) +
  default.theme
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

```
#5 Adding all months as levels

unique(PeterPaul.chemistry$month)
```

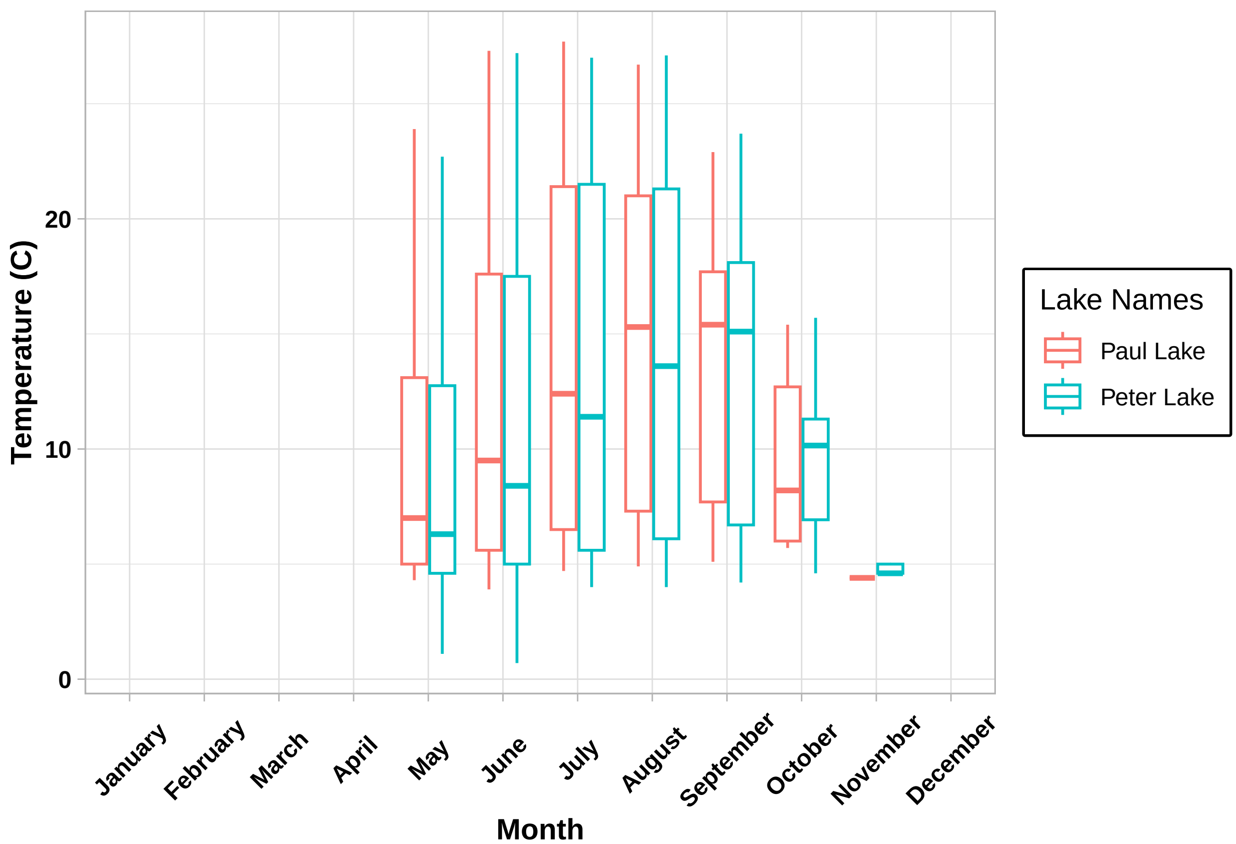
```
## [1]  5  6  7  8  9 10 11  2
```

```
PeterPaul.chemistry$month <- factor(PeterPaul.chemistry$month)

levels(PeterPaul.chemistry$month) <- c(
  levels(PeterPaul.chemistry$month),"1","3","4","12")

PeterPaul.chemistry$month <- factor(
```

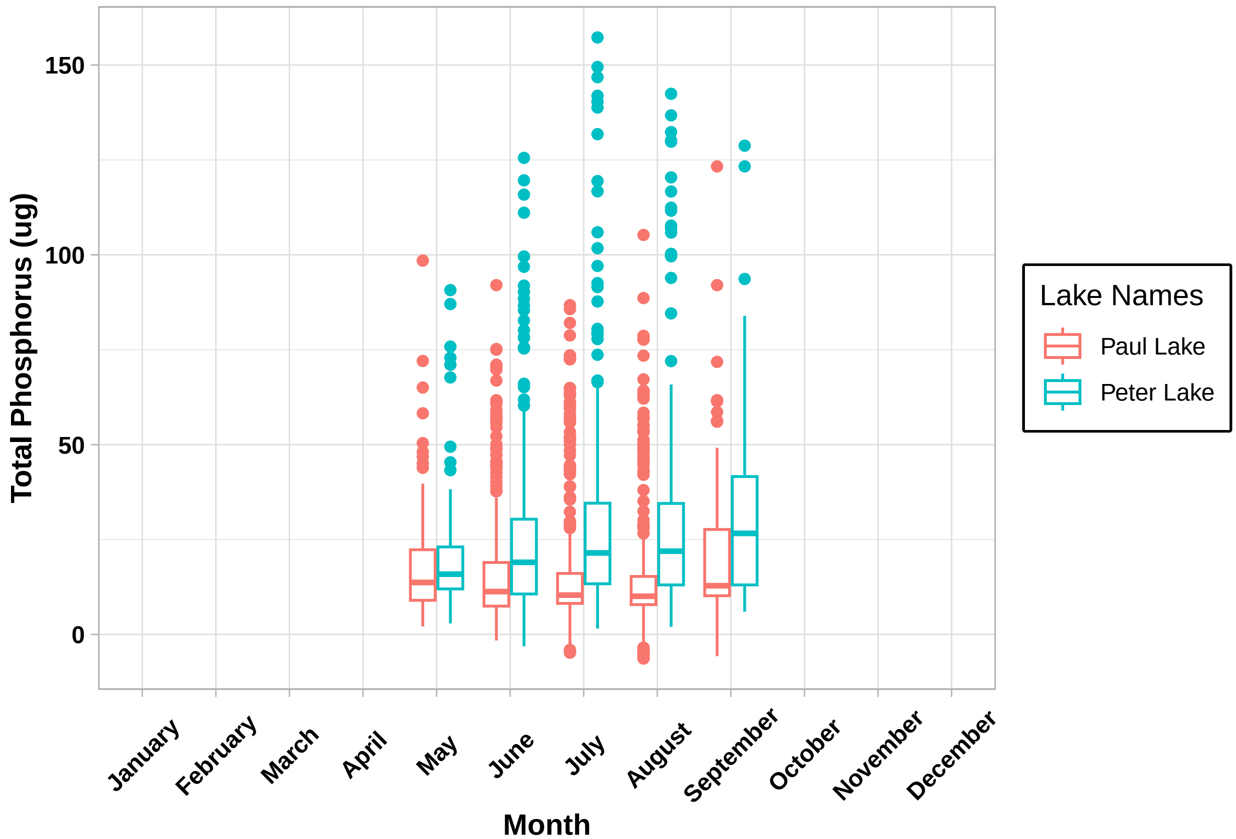
```r
  PeterPaul.chemistry$month,
  levels=c("1","2","3","4","5","6","7","8","9","10","11","12"))

PeterPaul.chemistry$month <- factor(
  PeterPaul.chemistry$month,
  levels=c("1","2","3","4","5","6","7","8","9","10","11","12"),
  labels=c(
    "January","February","March","April","May","June",
    "July","August","September","October","November","December"))

levels(PeterPaul.chemistry$month)
```

```
##  [1] "January"   "February"  "March"     "April"     "May"       "June"
##  [7] "July"      "August"    "September" "October"   "November"  "December"
```

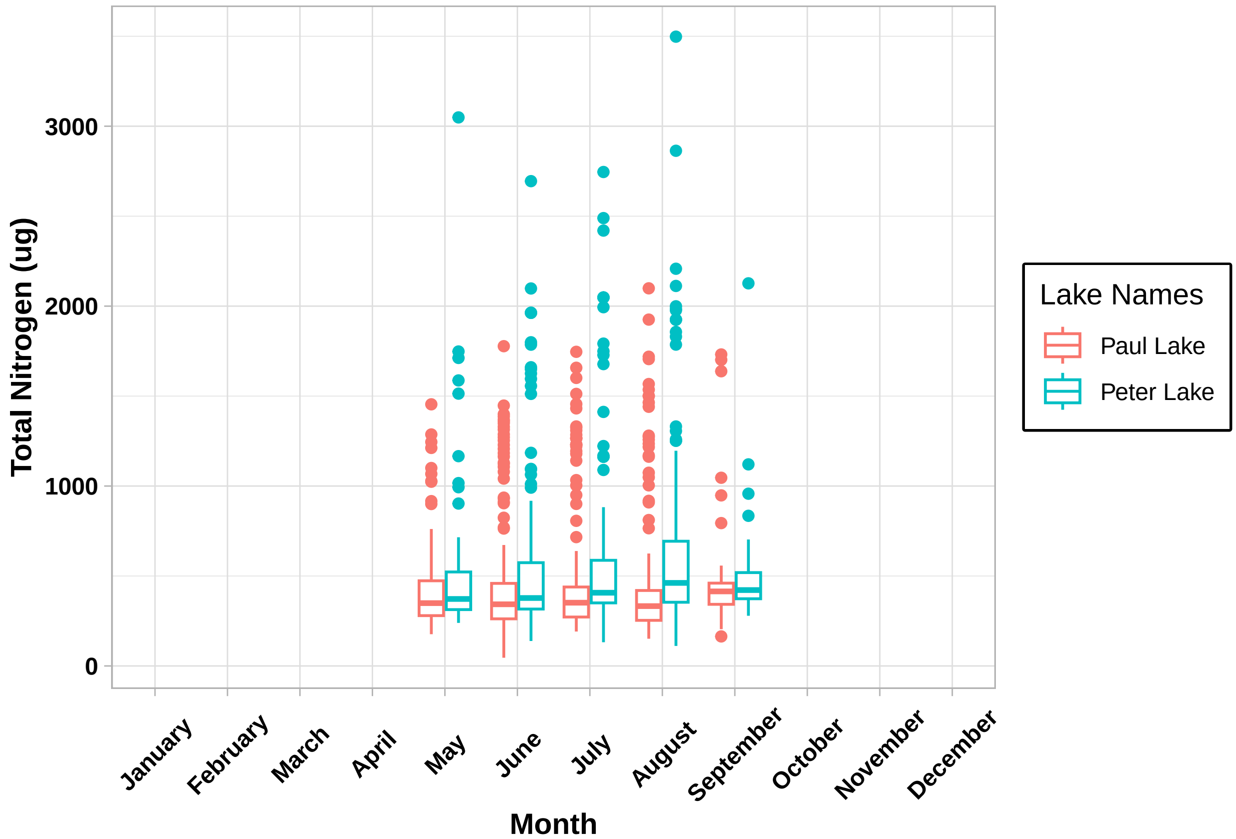
```r
# a) Boxplot for Temperature
Temperature_box <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = temperature_C, color = lakename)) +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Temperature (C)",color = "Lake Names") +
  default.theme +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5))

print(Temperature_box)
```



```r
# b) Boxplot for TP
Phosphorus_box <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = tp_ug, color = lakename))  +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Total Phosphorus (ug)",color = "Lake Names") +
  default.theme +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5))

print(Phosphorus_box)
```

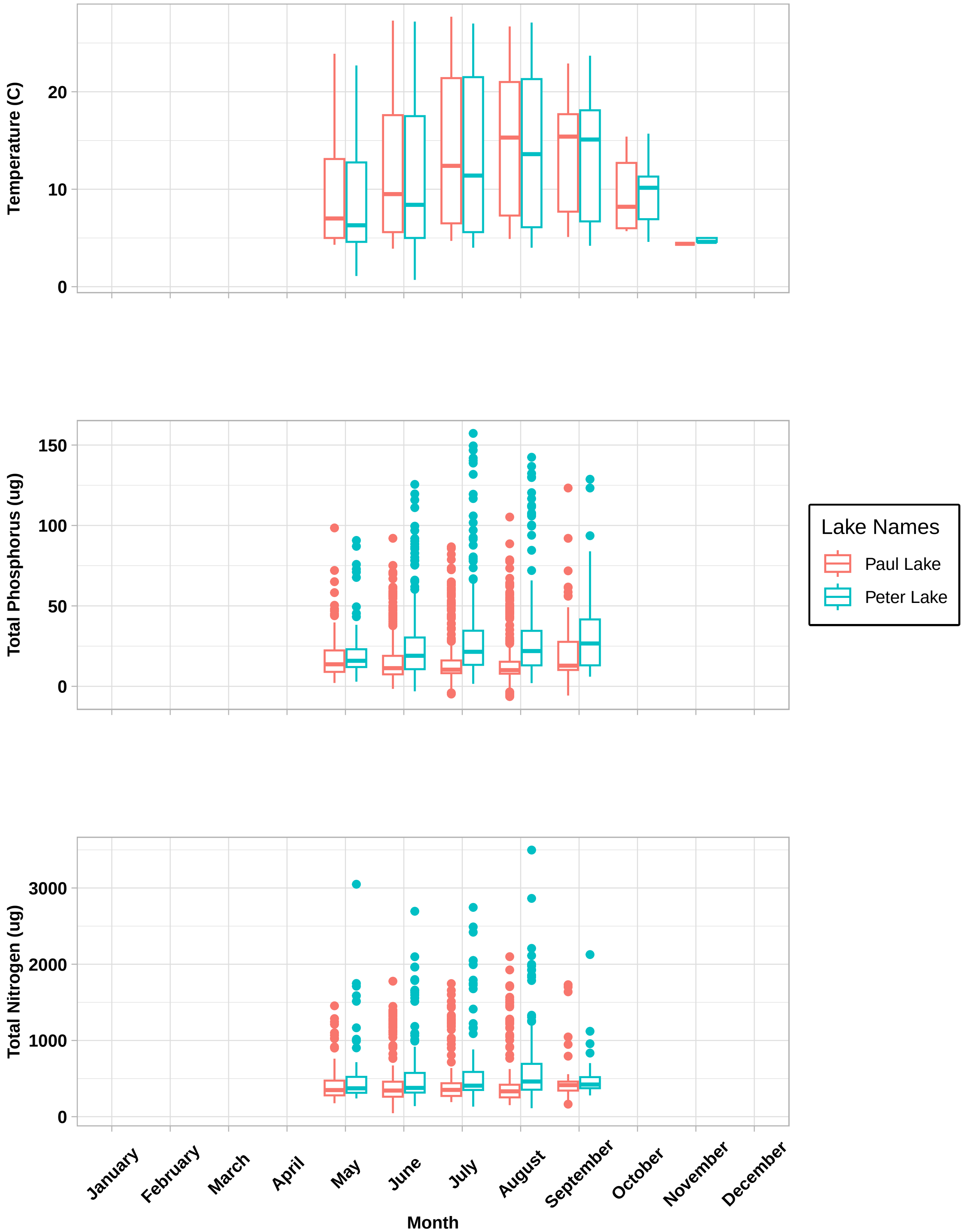



```r
# c) Boxplot for TN
Nitrogen_box <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = tn_ug, color = lakename))  +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Total Nitrogen (ug)",color = "Lake Names") +
  default.theme +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5))

print(Nitrogen_box)
```




```
# 5 d) Cowplot combining all plots
Temperature_cowplot <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = temperature_C, color = lakename)) +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Temperature (C)",color = "Lake Names") +
  default.theme +
  theme(
    axis.title = element_text(size = 9),
    axis.title.x = element_blank(),
    axis.text.x = element_blank(), legend.position = "none")

Phosphorus_cowplot <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = tp_ug, color = lakename))  +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Total Phosphorus (ug)",color = "Lake Names") +
  default.theme +
  theme(
    axis.title = element_text(size = 9),
    axis.title.x = element_blank(),
    axis.text.x = element_blank())

Nitrogen_cowplot <- ggplot(PeterPaul.chemistry) +
  geom_boxplot(aes(x = month, y = tn_ug, color = lakename))  +
  scale_x_discrete(drop=FALSE) +
  labs(x="Month",y="Total Nitrogen (ug)",color = "Lake Names") +
  default.theme +
```

```
  theme(
    axis.title = element_text(size = 9),
    axis.text.x = element_text(
      angle = 45, vjust = 0.5), legend.position = "none")

plot_grid(
  Temperature_cowplot, Phosphorus_cowplot, Nitrogen_cowplot,
  align = 'vh', ncol = 1)
```

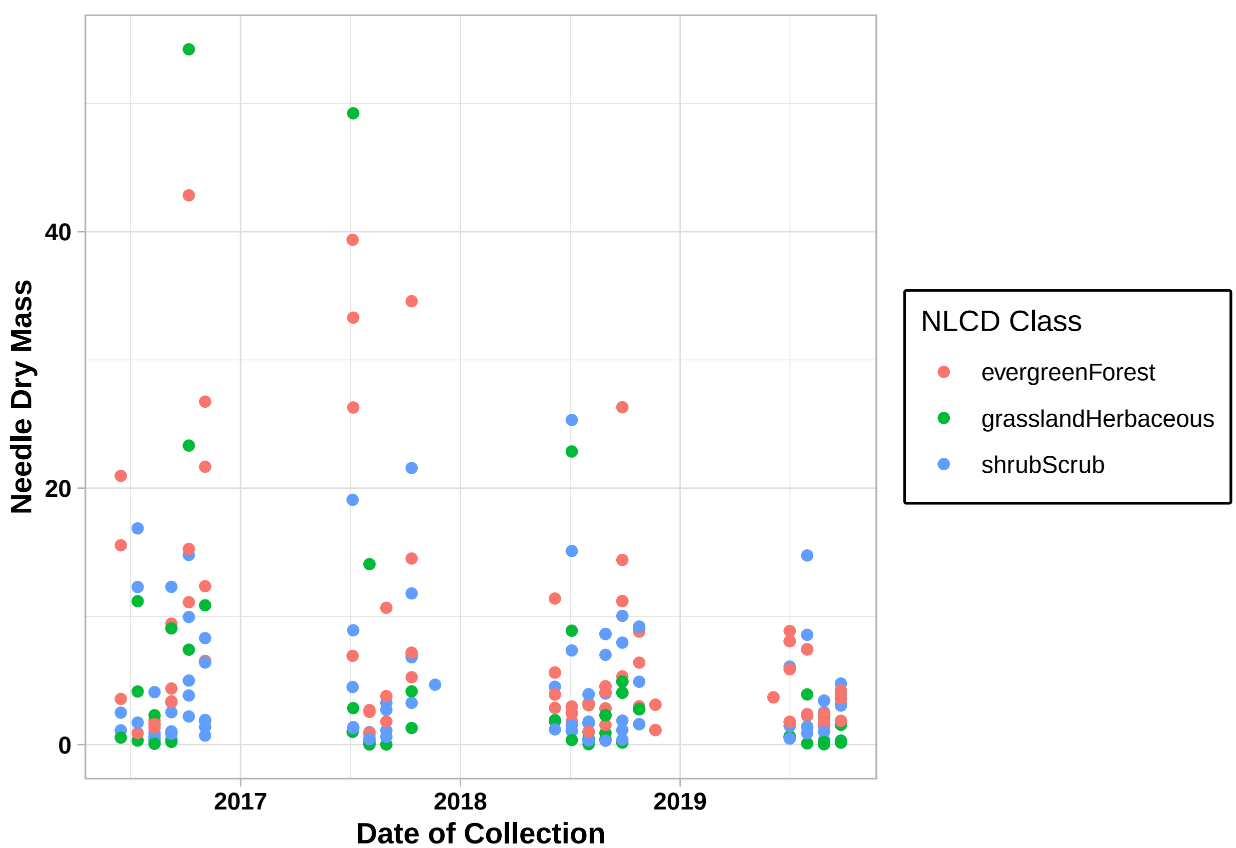



Question: What do you observe about the variables of interest over seasons and between lakes?

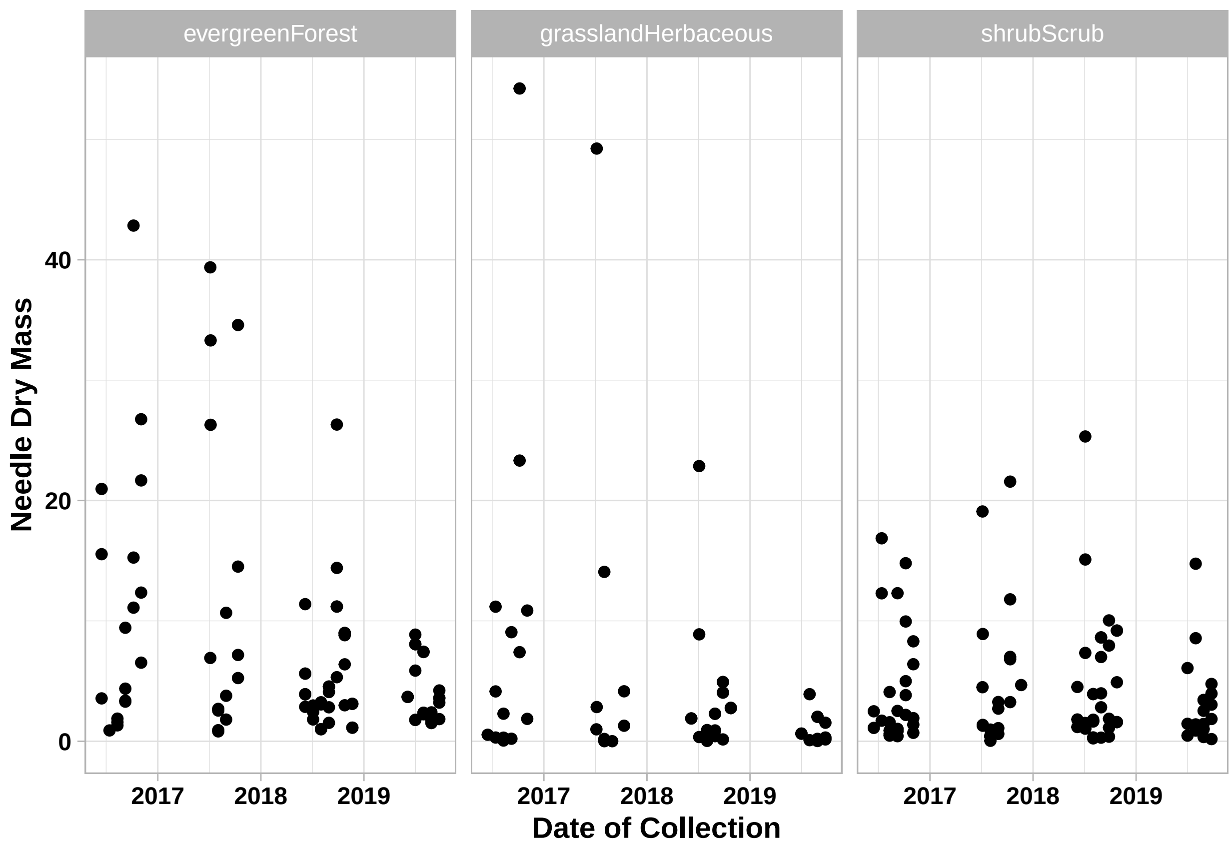
Answer: Given the monthly limitations of data, it is difficult to estimate how the concentrations of nitrogen and phosphorus and temperatures vary for Peter and Paul lakes during the colder months (December-March). That aside, the concentration of nitrogen in both lakes is much higher compared to phosphorus. The concentrations of both phosphorus and nitrogen seem to be increasing with a rise in temperature. However, the change in nitrogen is not as significant as phosphorus, and phosphorus concentrations seem to be rising much more in Peter Lake compared to Paul Lake. It cannot be estimated if the dip in temperature in October and November also coincides with decreasing nutrient concentrations.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6

Litter.needles <-
  ggplot(subset(Litter, functionalGroup == "Needles"),
         aes(x = collectDate, y = dryMass, color = nlcdClass)) +
  labs(x="Date of Collection",y="Needle Dry Mass",color = "NLCD Class") +
  geom_point() +
  default.theme

print(Litter.needles)
```



```r
#7

Litter.faceted <-
  ggplot(subset(Litter, functionalGroup == "Needles"),
         aes(x = collectDate, y = dryMass)) +
  labs(x="Date of Collection",y="Needle Dry Mass") +
  geom_point() +
  facet_wrap(vars(nlcdClass), ncol = 3) +
  default.theme

print(Litter.faceted)
```




Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: I think plot 7, where the classes are faceted, is more effective. This is because the plot points in 6 are overlapping in several places, making it difficult to identify the dry mass for each class. In plot 7, we can see a clearer distribution of all the plot points and how the classes differ in dry mass.